

AI foundation syllabus

Materials: [AI_foundation_tutorial](#), Prepare: [1_AX_trend](#)

Category	Subject	Reference	Weight	Memo
AX Trends and Services	AX Trends		0.5	Theory
	Comparison of AI vs. General Software Architecture			
	AI Service Development Flow			
Fundamental Concepts of Machine Learning	AI Learning and Development Environment	2_ML_basic	0.5	Example
	Difference between Classification and Regression			
	Concepts of Overfitting, Regularization, Validation Set			
	Pipeline using Scikit-learn			
Structural Understanding of Deep Learning	Basic Neural Network Structure (Perceptron, MLP)	3_DL_foundation	1.5	Theory
	Meaning of Optimizer, Learning Rate, Batch Size			
	Conceptual Differences of CNN/RNN/LSTM			
	Tensorflow vs. Pytorch Comparison			
Basic Flow of Natural Language Processing	Text Preprocessing	4_NLP	1.5	Example
	Tokens and Word Embedding			
	Introduction to Hugging Face Model Examples			
Transformer and AI Models	Understanding the Transformer Structure (Self-Attention, Positional Encoding)	5_transformer	1	Theory
	Summary of the Operating Principles of BERT, GPT, etc.			

	Why has the Transformer become the standard?			
LLM and AI Agent	LLM and RAG	6_LLM_agent_vibe	1.5	Example
	Open Source and Ollama			
	AI Agent			
	Generative AI and Vibe Coding			
Overview of AI Model Servicing	Model Training vs. Serving	7_service	1	Theory
	Introduction to API-based AI Service Architecture			
	Overview of Light-weighting and Deployment Strategies (ONNX, FastAPI, Flask, etc.)			
AX Strategy	Considerations for Corporate AI Adoption	8_AX_reference	0.5	Theory
	Q&A and Recommended Learning Materials			

* Note: The detailed order and content of each unit may be partially modified for smooth delivery.